

Statistics Mechanics Notes

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Contents

1	Information Theory	2
1.1	Information	2
1.2	Information Entropy	3
2	Manifold	4
2.1	Basic Concept of Manifold	4
2.2	Differential Form	5
2.3	Legendre Transformation	6
3	Thermodynamics	7
3.1	Basic Thermodynamics	7
3.2	Maxwell Relation	8
3.3	Information Theory Based Derivation	9
4	Classical Statistical Mechanics	10
4.1	Microcanonical Ensemble	10
4.2	Canonical Ensemble	11
4.3	Grand Canonical Ensemble	12
4.4	NPT Ensemble	12
5	Cluster Expansion	13

1 Information Theory

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In this section, I will introduce the information theory in a very non-rigorous way, to make it easier for understanding.

1.1 Information

Before introducing the concept of information, we have to give some reflection based on daily experience. What does the word “information” really mean? If you already know that there is raining outside, will you gain more information if others tell you the weather is rainy? Of course not. This example indicates that, information is something that related to the uncertainty. If you gain some knowledge that is already known, that will not give any extra information since there’s no uncertainty. There is a powerful mathematical tool that can quantify the uncertainty, which is called probability. Now we have a general feeling that, if we want to quantify the information, we have to formulate it with probability. Here we can denote it as $I[P(x)]$, or $I(x)$, where the x is the observation value.

Let’s consider another case. For two totally independent events, like “tomorrow will be rainy” and “the sun is in galaxy”. By knowing these two events, the information given by these are totally additive. That means, knowing one event will not influence the information of knowing another event. Formulate it in more rigorous language, that is to say, for two totally independent events, we have equation:

$$I(x, y) = I(x) + I(y) \quad (1.1)$$

A natural way to realize this is the logarithm function. We can define the information as:

$$I(x) = -k \ln P(x) \quad (1.2)$$

Information will be positive when k is positive. Thus, we get the expression for information. We can treat this quantity as the surprise when you receive the occurrence of certain event. If an event that is very rare to happen, then knowing that event will give you a lot of information. In computer science or communication, the definition will usually be written as:

$$I(x) = -\log_2 P(x) \quad (1.3)$$

Which can be directly interpreted into the unit of bits. The base of 2 refers to the 0 and 1 in binary case. For a 50–50 percent probability case, it will carry 1 bit information, or means we only need 1 time of binary measure we can pinpoint the reality. A typical application for this definition is that, if there are 1023 bottles of water, and 1 bottle of

saline. How much binary test we need to take under the worst case? The answer is 10 because:

$$-\log_2 \frac{1}{1024} = 10 \tag{1.4}$$

This event carries 10 bits of information.

1.2 Information Entropy

Since information is a quantity that related with probability, it's natural to think that, what the expectation value of information means. Here, for a random variable X , we define a new quantity, $S(X)$, as:

$$S(X) = - \sum_x k P(x) \ln P(x) = \langle I \rangle \tag{1.5}$$

Which is called information entropy or Shannon entropy. This quantity is used to quantify the average information of a class of uncertain events.

2 Manifold

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2.1 Basic Concept of Manifold

Here we only briefly introduce the concept of manifold, and some basic calculation rules, which is useful for the following derivations. The n dimensional manifold M is an object that is continuous and smooth. At each local points, it looks like Euclidean space $\mathbb{R}^n$. For example, the surface of a sphere is a 2-manifold. For each certain points, if we zoom in, the region around that point will look like a 2D plane. Just like the earth. We feel the ground where we stand at is a plane, while from a macroscopic view, the earth is a sphere.

Manifold itself is an abstract object. To describe the manifold, usually we need to define a set of coordinate functions. For n -manifold, if there exists a set of continuous functions $\{X^1, X^2, \dots, X^n\}$, that for each point p in the manifold, there is a map

$$(X^1(p), X^2(p), \dots, X^n(p)) \in U \subset \mathbb{R}^n. \quad (2.1)$$

Where U is a set of n dimensional continuous region in the $\mathbb{R}^n$ space. More intuitively speaking, n -manifold is isomorphic with a set of n dimensional continuous region in the $\mathbb{R}^n$, and coordinate functions are the one to one mapping between these two objects.

After define a set of coordinate functions, it will be easier for us to do research on other continuous differentiable functions. Assuming we have a continuous function f , that for each point p on the manifold, we have $f(p) \rightarrow \mathbb{R}$. Since the manifold is locally equivalent to a set of continuous region in $\mathbb{R}^n$, we can try to pretend that the continuous region in $\mathbb{R}^n$ is the manifold itself. For each point p , it has a tangent space and cotangent space, which constitutes the dual space. In order to reduce the abstract terminology, we can go back to the picture of the surface of a sphere, or earth. The tangent space is the space that tangent to the surface at p . Imagine you stand on the ground, and you can't fly. If you want to move, you have to select a direction. All the direction you can choose is limited into a 2D plane. That's the tangent plane. Let's back to the n -manifold. At each point p , the tangent space can be represented under the basis $\{\frac{\partial}{\partial X^i}\}$.

Now we get the tangent space done. Next let's look into cotangent space. Note that, here the cotangent is not the real cotangent, it's just the dual space of the tangent space, which is not the cotangent in the geometry. This dual space is designed for quantifying the difference of a certain function value at p . For example, you can define a function of elevation at each point on earth, or define a function of magnetic intensity. It's assumed that, when you make a very small movement, which causes a small changes in the coordinates. The relation between small difference of the function and the small movement can be treated as linear. For a general case, it means that, for the continuous differentiable function f , the small difference of the function df , can be expressed by the basis of dX^i . That means:

$$df = a_i dX^i \quad (2.2)$$

Since there are two spaces that constitute the dual relation, we have to define that how the dual space interact with each other. The tangent space provides the direction which you want to explore, while the cotangent space provides the quantity you want to evaluate. The interaction between the dual space and space is that, for any continuous differentiable function f , the operation of the cotangent space element df and tangent space element $\frac{\partial}{\partial X^j}$ is:

$$df\left(\frac{\partial}{\partial X^j}\right) = \frac{\partial f}{\partial X^j} \quad (2.3)$$

If $f = X^i$, we define:

$$dX^i\left(\frac{\partial}{\partial X^j}\right) = \frac{\partial X^i}{\partial X^j} = \delta_j^i \quad (2.4)$$

It's trivial because the coordinate function will only change when you move along that coordinate. Combining the previous definition, we have:

$$df\left(\frac{\partial}{\partial X^j}\right) = a_i dX^i\left(\frac{\partial}{\partial X^j}\right) = a_i \delta_j^i \quad (2.5)$$

Thus, the coefficient a_i , which is the component of df on the basis dX^i , is $\frac{\partial f}{\partial X^i}$.

2.2 Differential Form

The notation d here undertake some specific physical meaning, like in calculus. Here we will still only give a straightforward picture, instead of rigorous definition or derivations. You can understand it as, d is an operation that, measure the current of a quantity at an infinitesimal boundary, maybe the object is a function, or field. For example, the elevation function g . When you walk along a small distance, dx , and at each point on the trajectory has a gradient. dg reflects the gradient accumulation along the dx , which is the difference of the elevation function. Or we can understand it conversely. The difference of the elevation function after you walked a small distance dx , is equal to the accumulation of the gradient at each point. Same meaning applies to d^2g , and so on. That means, the difference of $(n-1)$ -form at the boundary, is equal to the accumulation of n -form within the area.

OK maybe it's too abstract, let's treat it as calculus, and remember some basic calculation rules. Here we need to define a rule called wedge product, which is:

$$da \wedge db \quad (2.6)$$

The wedge product you can treat it as the area that you learned during the calculus, $dx dy$, which is an infinitesimal area. But there's a bit of difference, that the $da \wedge db$ contains the information of direction. Here we have some basic calculation rules for wedge product, assume da, db, dc are 1-form:

$$d(a + b) \wedge dc = da \wedge dc + db \wedge dc \quad (2.7)$$

$$da \wedge db = -db \wedge da \quad (2.8)$$

$$d(ab) = (da)b + a(db) \quad (2.9)$$

In general, it's very similar to the case in calculus, only note that we need to add a minus sign if we swap the order. It's trivial that we can infer $da \wedge da = 0$. Note that, if the basis of the 1-form df can be denoted as dX^i , then the basis of the 2-form can be denoted as $dX^i \wedge dX^j$, where $i \neq j$. If a is a smooth 0-form, then we have $d^2a = 0$.

2.3 Legendre Transformation

Legendre transformation is a technical skill that applies to the manifold. By which we can change the coordinate function. For example, assume we have a function f , with 1-form relation:

$$df = ydx \quad (2.10)$$

Here we can treat x is the natural variable of function f , which is $f(x)$. Then we can deduct the $d(xy)$ in both side, which has:

$$df - d(xy) = y dx - d(xy) = -x dy \quad (2.11)$$

If we define $g = f - xy$, then we have:

$$dg = -x dy \quad (2.12)$$

And the function y is the natural variable of function g .

3 Thermodynamics

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In this section we are going to introduce the basic thermodynamics. We will first, go over the basic thermodynamics from a more abstract point of view, manifold, to understand the mathematical structure within the thermodynamics. Later, we will try to re-formalize it by the tools of information theory, revealing the statistics nature of thermodynamics.

3.1 Basic Thermodynamics

Let's go over the basic assumption of thermodynamics. There are four laws that constitute the fundamental of thermodynamics.

The zero law is that: If there are three systems, A , B and C . If A and B reach the thermal equilibrium, B and C reach the thermal equilibrium, then A and C will definitely reach the thermal equilibrium. This law ensures that the definition of the temperature.

Assume we have an enclosed system. The first law can be expressed as:

$$dU = dQ - dW \quad (3.1)$$

Here Q is the heat interaction energy, and W is the mechanical work. This denotes the energy conservation, that the total energy change is equal to the summation of the contribution from the heat exchange and the mechanical work.

The second law can be expressed as:

$$\Delta S \geq \frac{\delta Q}{T}. \quad (3.2)$$

This inequality indicates that, the entropy of an isolated system will always increase, since $\delta Q = 0$.

The third law is that, when $T \rightarrow 0$, $S \rightarrow 0$ for perfect crystal with individual ground state.

Now let's look into it more abstractly. The equilibrium state of a thermodynamics system, can be treated as an abstract 2D manifold. Here we only consider the fixed particle number case. According to the first law, we can give out the 1-form on the manifold, that is:

$$dU = T dS - p dV \quad (3.3)$$

Here we can see that, S and V are the natural coordinate function defined on the manifold. Thus for the function U , it's natural to denote it as $U(S, V)$. We can apply Legendre

transformation on this equation, to change the coordinate functions to its dual case. That is:

$$dU - d(TS) = -S dT - p dV \quad (3.4)$$

$$dU + d(pV) = T dS + V dp \quad (3.5)$$

$$dU - d(TS) + d(pV) = -S dT + V dp \quad (3.6)$$

Thus, we can define another three different potentials as:

$$F(T, V) = U - TS \quad (3.7)$$

$$G(T, p) = U - TS + pV \quad (3.8)$$

$$H(S, p) = U + pV \quad (3.9)$$

H is the enthalpy, G is the Gibbs free energy, F is the Helmholtz free energy.

3.2 Maxwell Relation

Once we accept the language of manifold, the Maxwell relation can be obtained very naturally. For the four different potentials, they are smooth scalar functions, in that case, the second exterior derivatives have to be zero. So we have:

$$\begin{aligned} 0 &= d(dU) = d(T dS - p dV) \\ &= \left(\frac{\partial T}{\partial V}\right)_S dV \wedge dS - \left(\frac{\partial p}{\partial S}\right)_V dS \wedge dV \\ &= -\left(\frac{\partial T}{\partial V}\right)_S dS \wedge dV - \left(\frac{\partial p}{\partial S}\right)_V dS \wedge dV \end{aligned} \quad (3.10)$$

The subscript of the bracket denotes which coordinate functions you are using, reflecting that you are moving along which coordinate function are fixed on the manifold. Similarly, we have:

$$\begin{aligned} 0 &= d(dF) = d(-S dT - p dV) \\ &= -\left(\frac{\partial S}{\partial V}\right)_T dV \wedge dT - \left(\frac{\partial p}{\partial T}\right)_V dT \wedge dV \\ &= \left(\frac{\partial S}{\partial V}\right)_T dT \wedge dV - \left(\frac{\partial p}{\partial T}\right)_V dT \wedge dV \end{aligned} \quad (3.11)$$

$$\begin{aligned}
0 &= d(dG) = d(-S dT + V dp) \\
&= -\left(\frac{\partial S}{\partial p}\right)_T dp \wedge dT + \left(\frac{\partial V}{\partial T}\right)_p dT \wedge dp \\
&= \left(\frac{\partial S}{\partial p}\right)_T dT \wedge dp + \left(\frac{\partial V}{\partial T}\right)_p dT \wedge dp
\end{aligned} \tag{3.12}$$

$$\begin{aligned}
0 &= d(dH) = d(T dS + V dp) \\
&= \left(\frac{\partial T}{\partial p}\right)_S dp \wedge dS + \left(\frac{\partial V}{\partial S}\right)_p dS \wedge dp \\
&= -\left(\frac{\partial T}{\partial p}\right)_S dS \wedge dp + \left(\frac{\partial V}{\partial S}\right)_p dS \wedge dp
\end{aligned} \tag{3.13}$$

That's the Maxwell relation. Note that, all the 1-form has the structure as:

$$dU = A da + B db + \dots \tag{3.14}$$

Can obtain the similar relation here, by applying the Legendre transformation and d . The Maxwell relations are precisely such integrability conditions for the thermodynamic potentials.

3.3 Information Theory Based Derivation

4 Classical Statistical Mechanics

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In this section, we will derive the probability distribution of different ensemble. Here “ensemble” means the system under different specific boundary conditions. Typically, there are microcanonical ensemble, canonical ensemble, grand canonical ensemble, and NPT ensemble. Microcanonical ensemble is the system that has the number of particles N , the total volume V , and the total energy E , fixed. Canonical ensemble is the system that has N , V fixed, while allowing exchange energy with surroundings. Grand canonical ensemble allows system to exchange not only energy, but the particles. Only requires V fixed. For NPT ensemble, it’s obviously that, it fixes the N , temperature T and pressure P . In general, the probability distribution for all kinds of ensemble, can be derived by maximizing the Shannon entropy, with constraints applied.

4.1 Microcanonical Ensemble

Let’s look into the easy case, the microcanonical ensemble. In microcanonical ensemble, The total number of particles N , total volume V , and total energy E , are fixed. Assume allowed specific configuration σ_i will appear at probability p_i , then the only constraints we can apply on the system is the normalization condition, which is:

$$\sum_i p_i = 1 \quad (4.1)$$

The Lagrange multiplies can be written as:

$$L = - \sum_i p_i \ln p_i - \alpha (\sum_i p_i - 1) \quad (4.2)$$

Taking derivative gives the equation:

$$\frac{\partial L}{\partial p_i} = -1 - \ln p_i - \alpha = 0 \quad (4.3)$$

We can immediately obtain $p_i = \text{const.}$ Assuming there’s W allowed configurations for given N , V , E , then the probability of any one specific configuration is:

$$p_i = \frac{1}{W} \quad (4.4)$$

And the entropy is:

$$S = -k_B \sum_i p_i \ln p_i = k_B \ln W \quad (4.5)$$

4.2 Canonical Ensemble

For the canonical ensemble, which fixes the particle number N , the volume V . The total energy for a specific configuration σ_i is no longer fixed, but we know the average energy U :

$$U = \sum_i E_i p_i = \langle E \rangle \quad (4.6)$$

Also, there is the probability normalization condition, then the Lagrange multiplies is:

$$L = - \sum_i p_i \ln p_i - \alpha (\sum_i p_i - 1) - \beta (\sum_i E_i p_i - U) \quad (4.7)$$

Taking the derivative gives the equation:

$$\frac{\partial L}{\partial p_i} = -1 - \ln p_i - \alpha - \beta E_i = 0 \quad (4.8)$$

Which gives:

$$p_i \propto e^{-\beta E_i} \quad (4.9)$$

We can set the normalization factor as $\frac{1}{Z}$, where Z is the partition function. Now we can get that:

$$p_i = \frac{e^{-\beta E_i}}{Z} \quad (4.10)$$

With:

$$Z = \sum_i e^{-\beta E_i} \quad (4.11)$$

Be careful, now the parameter β is just a Lagrange multiplier. And we haven't proven that it is the signal that commonly used in physics, $\frac{1}{k_B T}$. To show this, we have to build the connection with thermodynamics. Insert the expression of probability into the information entropy, we can get:

$$\begin{aligned} \frac{S}{k_B} &= - \sum_i p_i (-\beta E_i - \ln Z) \\ &= \beta \sum_i p_i E_i + \ln Z \\ &= \beta U + \ln Z \end{aligned} \quad (4.12)$$

So that we have:

$$\frac{1}{k_B} dS = \beta dU + U d\beta + \frac{\partial \ln Z}{\partial \beta} d\beta \quad (4.13)$$

Also:

$$\frac{\partial \ln Z}{\partial \beta} = \frac{1}{Z} \left(- \sum_i e^{-\beta E_i} E_i \right) = -U \quad (4.14)$$

Combine this two equations, we have:

$$dS = k_B \beta dU \quad (4.15)$$

Which gives $\frac{\partial S}{\partial U}|_{N,V} = k_B \beta$. For the canonical ensemble, the equilibrium state satisfies the deferential relation:

$$dU = T dS - p dV \quad (4.16)$$

Considering the volume is fixed, so $dV = 0$, which gives:

$$dU = T dS \quad (4.17)$$

It naturally gives that $\frac{\partial S}{\partial U}|_{N,V} = \frac{1}{T}$. So we have:

$$\beta = \frac{1}{k_B T} \quad (4.18)$$

4.3 Grand Canonical Ensemble

4.4 NPT Ensemble

5 Cluster Expansion

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The cluster expansion is an approach used to construct a basis set. More precisely, it is discrete Fourier transformation on the configuration space. On the basis, we can expand any lattice dependent only functions. It's a powerful tool for the investigation of atomic ordering.

Assuming we have a configuration space, Ω . The single site in the configuration space is denoted as σ_i , and the configuration space is $(\sigma_1, \sigma_2, \dots, \sigma_N)$. For each site, it has different occupation element, that is $A = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$.

To define the orthonormal basis here, we have to give out the definition of the inner product first. The inner product on the single site σ_i is defined as:

$$\langle f|g \rangle = \sum_{a=1}^n \nu_a f(a)g(a) \quad (5.1)$$

Here ν is a reference distribution, which gives the probability of each specific element occupy this site. The orthonormal basis is a set of basis that satisfy:

$$\langle \phi_m | \phi_n \rangle = \delta_{mn} \quad (5.2)$$

It can be proved that, the number of the basis function is equal to the total number of the species. The first basis function will always be constant:

$$\phi_0 = 1 \quad (5.3)$$

The many body basis is defined as:

$$\Psi_{S,r}(\sigma) = \prod_{i \in S} \phi_{r_i}(\sigma_i) \quad (5.4)$$

S is the set of the site index in this many body term. $r_i \in \{1, 2, \dots, n-1\}$.